

Praneeth Kumar Thummalapalli

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EDUCATION

New York University | Master of Science in Computer Science GPA: 3.85/4.0

Graduation Date: May 2025

B.N.M Institute of Technology | B.E (with honors) in Computer Science Engineering CGPA: 9.50

Graduation Date: June 2023

TECHNICAL SKILLS

- **Front-End:** HTML, ReactJS, React Native, CSS, Bootstrap
- **Programming Languages:** Java, Python, C, C++, JavaScript, SQL, GraphQL
- **Database:** MySQL, DBMS, MongoDB, DynamoDB, FireStore
- **Backend:** Django, Express.js, Node.js, Springboot, Maven, Flask
- **Cloud Technologies:** AWS (EC2, S3, DynamoDB, Amplify, API Gateway, Lambda, AppSync, Q, GCP(Firebase, VertexAI, BigQuery)
- **Modules:** NumPy, Sklearn, Pandas, PyTorch, Altair, Yolo, Vision Transformer
- **AI :** Linear regression, Logis Regression, SVM, K-Means, Principal Componet Analysis, Neural Networks, Transformers
- **Tools:** Git, Hadoop, Apache Spark, PowerBI, Dataiku

EXPERIENCE

Louis Vuitton, New York | Software Intern - Data Science Team

June 2024 - Current

- Engineered and automated an efficient pipeline for optimal customer-targeting utilizing Dataiku, using various data parameters from BigQuery
- Optimized data handling to address timeout issues and multi-user access to data processing pipeline, enabling simultaneous processing of tens of millions of data rows

New York University | Lead Software Developer [Android: [FireBeats](#)] [iOS: [FireBeats](#)] [[Product](#)]

Sept 2023 – June2024

- Orchestrated React Native AI-driven Remote Health Monitoring System development, securing NSF funding, advancing to NYU-Berkley Entrepreneurship Challenge semifinals and NYU Stern accelerator program with total of \$110,000 non-equity funding
- Integrated data from Fitbit, Garmin, Google-Fit, and Apple-HealthKit, streamlining data through GCP Functions to trigger AI models deployed on VertexAI, boasting over 95% accuracy in detecting atrial-fibrillation, hypertension, and sleep-apnea

SIAMAF HealthCare, incubated in IISC Bangalore, India | Research Intern

July 2022 – Dec 2022

- Built AI-model using texture-based classification, boosting breast cancer diagnosis accuracy by 15%
- Applied Transfer Learning with Resnet-18, achieving 99% accuracy and reducing training-time

Appzera, Bangalore, India | Software Developer Intern

Sept 2020 - July 2021

- Consolidated the data and presentation layers of the Bullet Train Project's web app, employing React components to enhance reusability and streamlined development; achieved a 30% increase in UI-UX responsiveness
- Optimized state management with Redux and Context-API, synchronizing data across components, reducing load-time by 30%

PATENT AND PUBLICATIONS

- **Plug and Play Cubicles (Funded by IEDC, Patent Pending; Patent No: 202341012174):** Scalable, plug-and-play home automation solution, employing a novel cubicle structure for integration of sensors
- **A Comparative Study of Machine Learning Techniques in Heart Disease Detection**, DOI: <https://doi.org/10.5281/zenodo.4515583>
- **AI for Remote Health Monitoring via Wearables:** Automated Identification of Atrial Fibrillation, Arrhythmia, Sleep Apnea, Hypertension
- **Cancer Cell Classification using Deep Learning** at the International Conference on Science, Technology and Engineering, 16 April 2023

CERTIFICATIONS

- Amazon Web Services Certified Solutions Architect – Associate
- Developer: Certified Kubernetes Application Developer (CKAD), Dataiku Developer, Data Analytics (IIT Roorkee)

PROJECTS

Improving Text-to-SQL Translation System using LLMs [Application: [Text-To-SQL](#)]

- Deployed an advanced React application to translate natural language instructions into executable SQL queries, integrating three different Large Language Models (LLMs) for enhanced accuracy and efficiency
- Leveraged the Spider dataset, containing over 10,000 questions and 5,693 unique SQL queries across 200 databases using Hadoop MapReduce, to benchmark system performance, achieving an execution accuracy of 70.4%

AI-Driven Nutritional Analysis Mobile Application [Android: [FireBeats](#)] [iOS: [FireBeats](#)]

- Implemented and deployed a system using YOLOv8 and GPT in a React Native mobile application to detect calories, fat, protein, and carbs from food photos captured on phone camera, automatically identifying food and portion sizes

NYC Property Insights: An interactive Geo-Spatial Visual Exploration by ZIPCODE [Application: [Property-insights](#)]

- Created a complex visualization tool analyzing over 300,000 data rows for NYC property insights using Altair, integrating datasets like pavement ratings, subway connectivity, geospatial data, NYPD incidents, and real estate sales, enhancing data-driven decision-making

Analysing Youth-Driven Drug Abuse Patterns in NYC

- Integrated recursive feature elimination and information-based feature selection techniques, optimizing hyperparameters with Grid-Search
- Discovered optimal results with a Random Forest model coupled with SelectKBest feature selector, achieving a remarkable accuracy of 91.25% and precision of 90.83%